MODULE 6 · LECTURE 27a

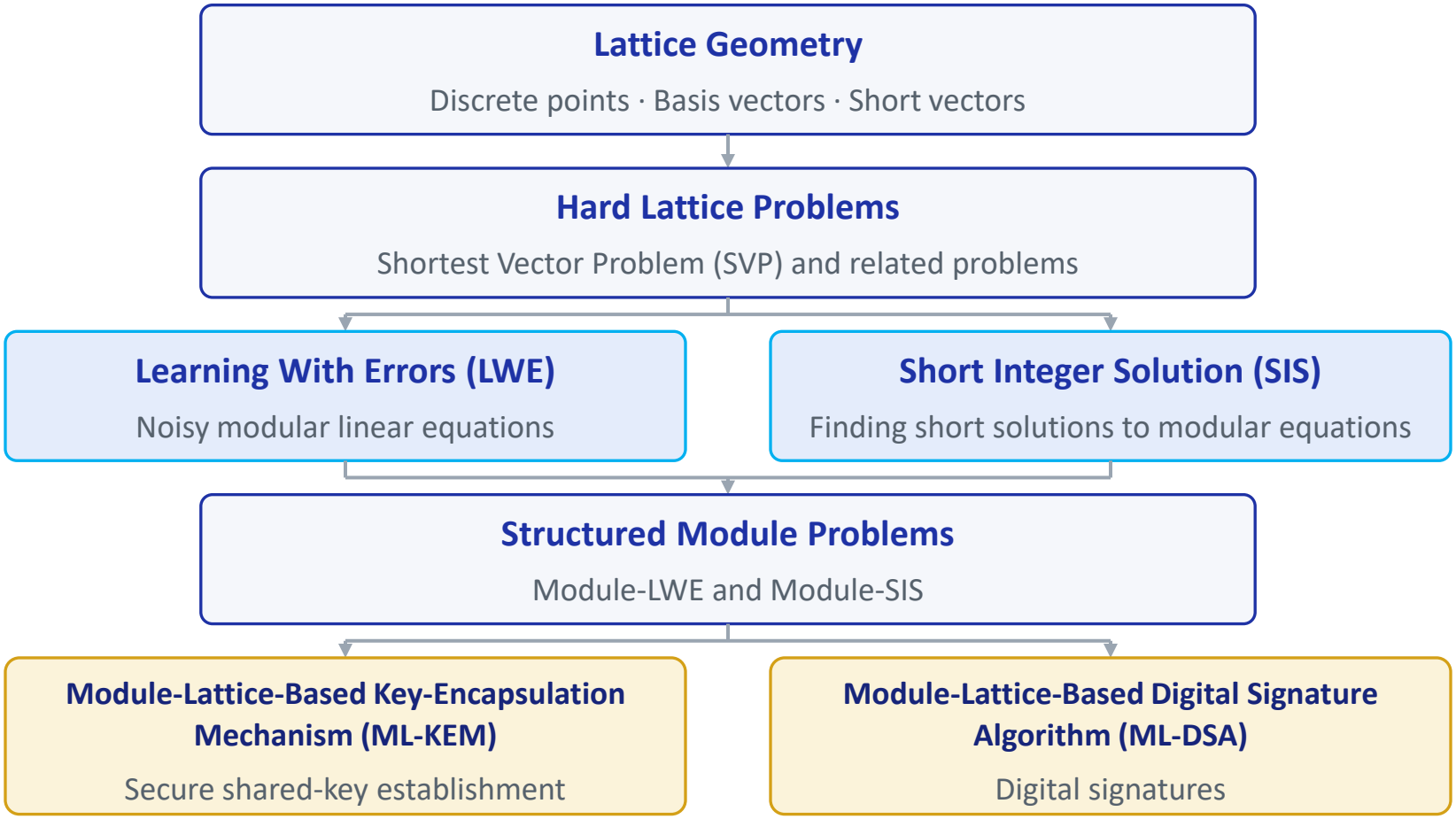
CS 6530 Applied Cryptography

Post-Quantum Cryptography Foundations

Module 6 – Post-Quantum Cryptography

From Lattice Geometry to Post-Quantum Cryptography

Lattice-based cryptography builds security on mathematical problems believed to be computationally difficult for both classical and quantum computers.



Conceptual roadmap; security depends on specific computational assumptions and parameter choices.

Acknowledgment: Prof. Alfred Menezes – Kyber and Dilithium - https://cryptography101.ca/kyber-dilithium/ Material

What Is a Lattice?

Definition:

A lattice is a discrete set of points generated by integer linear combinations of linearly independent basis vectors.

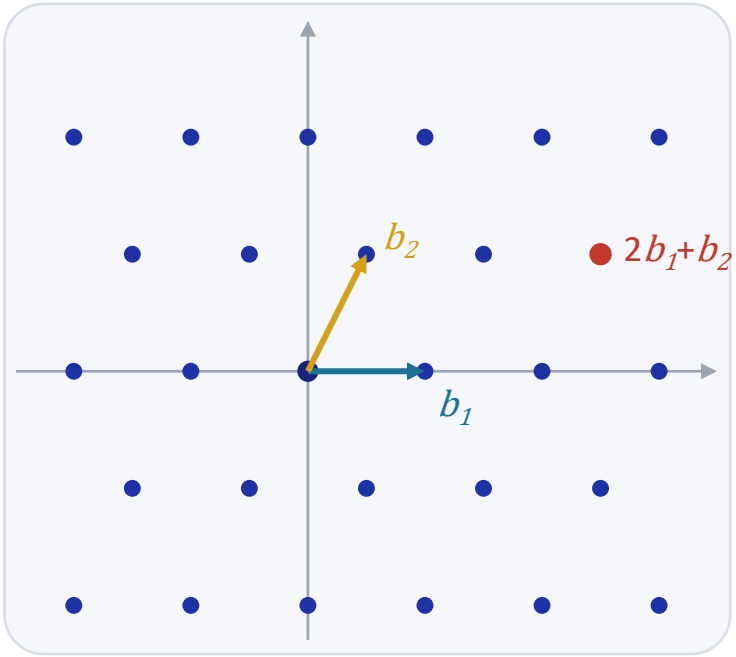
Mathematical expression:

$$L(B) = \{ \Sigma_{i=1}^{n} z_i b_i \mid z_i \in \mathbb{Z} \}$$

where $B = [b_1, b_2, \ldots, b_n]$ is a basis.

Basis Vectors

- Basis vectors are linearly independent.
- Coefficients are integers, not arbitrary real numbers.
- Integer combinations generate the lattice points.



Basis, Bases, and Basis Vectors

Term	Meaning
Basis vector	One of the linearly independent vectors used to generate a lattice through integer combinations.
Basis (singular)	A set of linearly independent vectors whose integer combinations generate every point of the lattice.
Bases (plural)	Different sets of basis vectors that can generate the same lattice.
Lattice	The set of all points generated by integer combinations of the basis vectors.

The Simplest Lattice — The Integer Grid

Consider two basis vectors:

$b_1 = (1,0), \quad b_2 = (0,1)$

Integer combinations generate:

$L_1 = \{ ab_1 + bb_2 \mid a,b \in \mathbb{Z} \}$

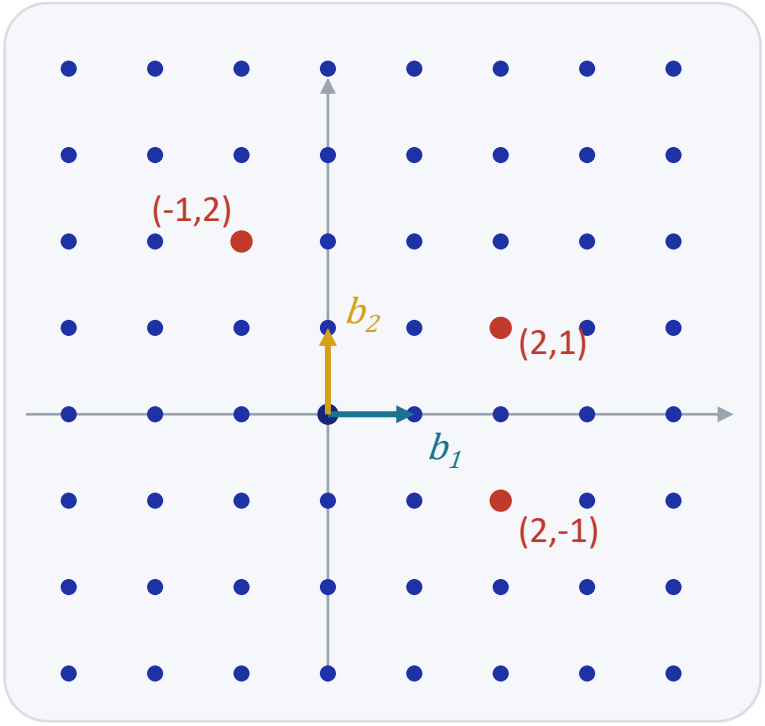
Examples:

(a,b)	Lattice Point
(1,0)	(1,0)
(0,1)	(0,1)
(1,1)	(1,1)
(2,1)	(2,1)
(2,-1)	(2,-1)
(-1,2)	(-1,2)

Therefore:

$$L_1 = \{(a,b) \mid a,b \in \mathbb{Z} \} = \mathbb{Z}^2$$

Every point with integer coordinates belongs to this lattice.



LATTICES · WORKED EXAMPLE

Constructing a Lattice from Basis Vectors

Given:

$b_1 = (2,0), \quad b_2 = (0,1)$

$L_2 = \{ ab_1 + bb_2 \mid a,b \in \mathbb{Z} \}$

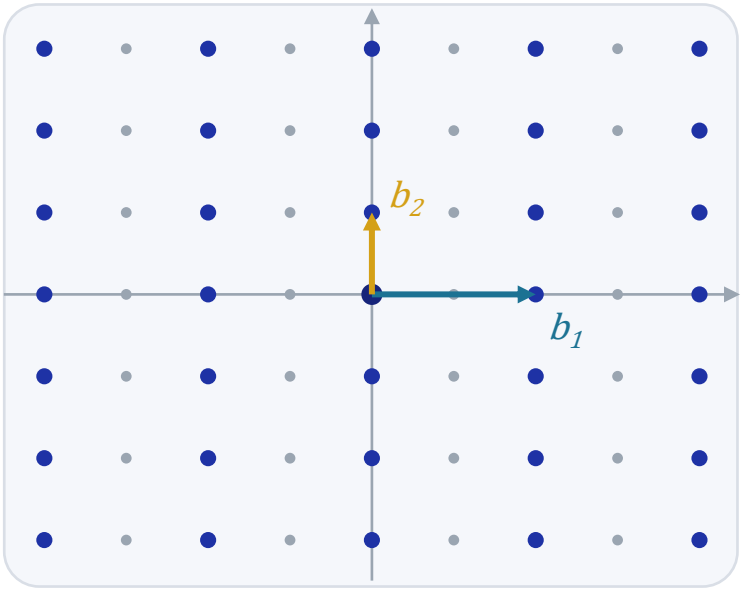
Examples:

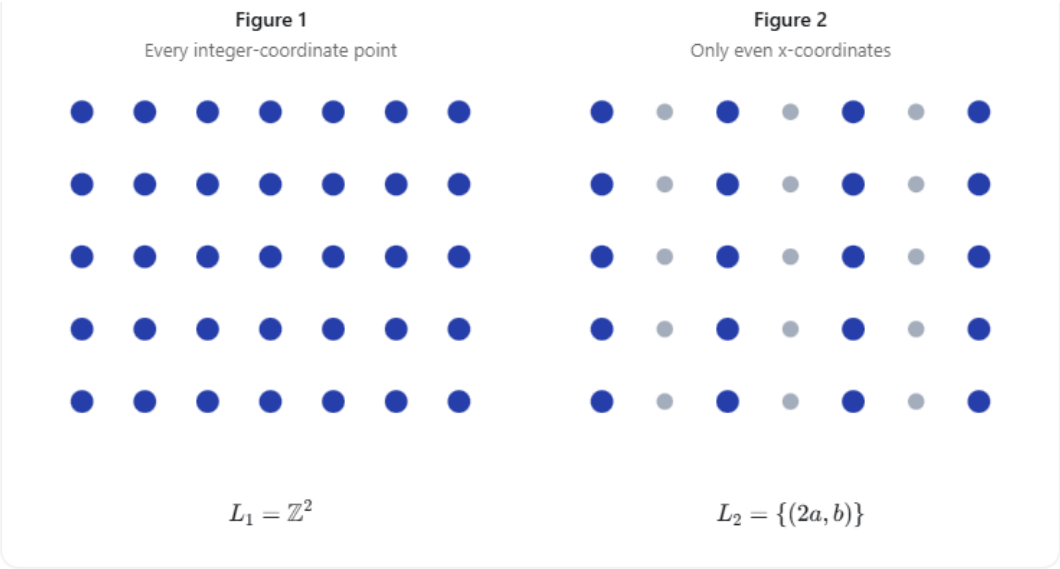
(a,b)	Lattice Point
(1,0)	(2,0)
(0,1)	(0,1)
(1,1)	(2,1)
(2,1)	(4,1)
(2,-1)	(4,-1)
(-1,2)	(-2,2)

Therefore:

$$L_2 = \{(2a,b) \mid a,b \in \mathbb{Z} \}$$

All lattice points have even x-coordinates — this lattice contains half the points of the integer grid ($\mathbb{Z}^2$).





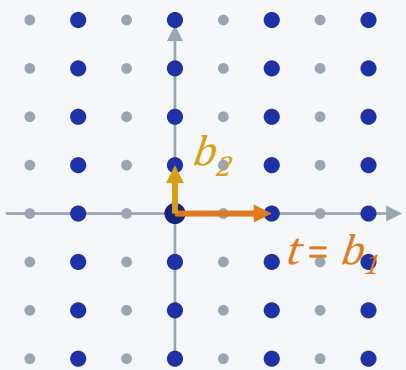
Different Bases, Same Lattice

A lattice can have different bases. Some bases are shorter and more nearly orthogonal, making lattice computations more convenient.

Recall the lattice: $L = \{(2a, b) \mid a, b \in \mathbb{Z}\}$.

Basis 1 — Short, orthogonal vectors

$b_1 = (2,0)$, $b_2 = (0,1)$

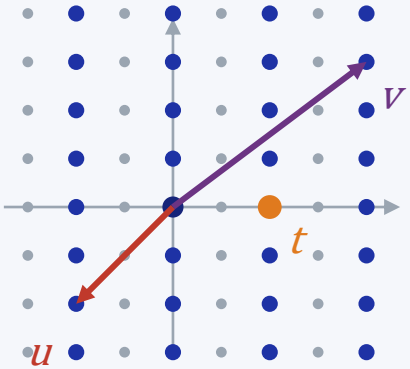


Using Basis 1:

$t = b_1$

Basis 2 — Longer, non-orthogonal vectors

$u = (-2,-2)$, $v = (4,3)$



Using Basis 2:

$t = 3u + 2v$

Verify: $3(-2,-2) + 2(4,3) = (-6,-6) + (8,6) = (2,0)$.

$u = -b_1 - 2b_2$, $v = 2b_1 + 3b_2$, $\det(U) = (-1)(3) - (2)(-2) = 1$

The change-of-basis matrix has determinant 1. Its inverse also has integer entries, so each basis can be expressed using integer combinations of the other. Hence both bases generate exactly the same lattice.

The Shortest Vector Problem (SVP)

Given a lattice basis, can we find the shortest nonzero lattice vector?

$$\lambda_1(L) = \min_{v \in L,\, v \neq 0} |v|_2$$

The Shortest Vector Problem (SVP) asks us to find a nonzero lattice vector whose Euclidean length is $\lambda_1(L)$.

Notation	Meaning
L	The lattice
$\mathbf{v} \in L$	Consider a vector belonging to the lattice
$\mathbf{v} \neq \mathbf{0}$	Exclude the zero vector
$\|\mathbf{v}\|_2$	Euclidean length of that vector
$\min$	Find the smallest such length
$\lambda_1(L)$	The length of a shortest nonzero lattice vector

Worked example:

$L = \{(2a,\ b) \mid a, b \in \mathbb{Z}\}$

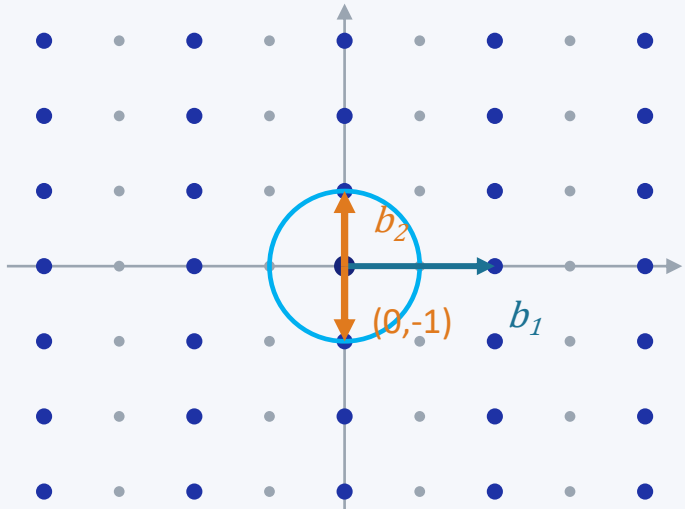
Two shortest: (0,1), (0,-1)

Length: $|(0,+1)|_2 = |(0,-1)|_2 = 1$

Compare: $|(2,0)|_2 = 2$

$$\lambda_1(L) = 1$$

Easy here; can be hard in high dimensions.



Circle = radius 1 about the origin. Shortest nonzero vectors (0,±1) highlighted in orange.

LATTICES · COMPUTATIONAL HARDNESS

From SVP to Lattice-Based Security

The computational difficulty of finding sufficiently short lattice vectors provides a foundation for lattice-based cryptography.

STAGE 1 — Lattice Problems. SVP: find a shortest nonzero lattice vector. Approximate SVP (approx-SVP): find a vector within an approximation factor of the shortest length.

STAGE 2 — Lattice-reduction Algorithms and Cryptanalysis. Lenstra–Lenstra–Lovász (LLL): polynomial-time reduction, weak guarantees in high dimensions. Block Korkine–Zolotarev (BKZ): stronger local reduction, trading effort for a better basis. Neither solves every hard instance exactly.

STAGE 3 — Cryptographic Hardness Assumptions.
Learning With Errors (LWE) and **Short Integer Solution (SIS)** are algebraic problems used to build lattice-based cryptography. LWE asks us to recover hidden information from noisy modular equations. SIS asks us to find a short, nonzero vector satisfying modular equations. Under specified conditions, their hardness is connected to difficult lattice problems, providing foundations for lattice-based cryptographic constructions.

Technical insight: worst-case, asymptotic results do not guarantee concrete security.

Modular Arithmetic — The Integers mod q

For an integer modulus $q \geq 2$:

$a \equiv b \pmod{q}$

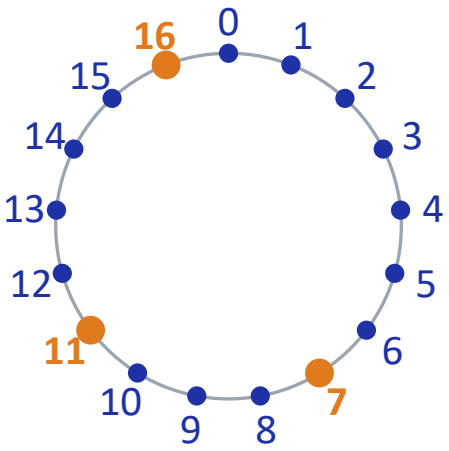
means that q divides (a − b).

The integers modulo q:

$$\mathbb{Z}_q = \{0, 1, \ldots, q-1\}$$

Addition, subtraction and multiplication are performed modulo q.

Worked example — q = 17:



9 + 15 = 24 ≡ **7** (mod 17)

9 − 15 = −6 ≡ **11** (mod 17)

9 × 15 = 135 ≡ **16** (mod 17)

Results wrap around modulo 17; highlighted residues 7, 11 and 16 are the computed answers.

Polynomials over $\mathbb{Z}_q[x]$

A polynomial in $\mathbb{Z}_q[x]$ has coefficients drawn from $\mathbb{Z}q$. Polynomial arithmetic is performed with coefficients reduced modulo q.

$\mathbb{Z}_q[x]$ = { polynomials in x with coefficients in $\mathbb{Z}_q$ }

Worked example — $q = 7$: $f(x) = 5 + 4x + 3x^2$, $g(x) = 6 + 3x + 2x^2$

term	const	x	x^2	x^3	x^4
f(x)	5	4	3	–	–
g(x)	6	3	2	–	–
f+g (raw)	11	7	5	–	–
f+g mod 7	**4**	**0**	**5**	**–**	**–**
f−g (raw)	−1	1	1	–	–
f−g mod 7	**6**	**1**	**1**	**–**	**–**

Coefficients above q are reduced modulo 7; the x-term in f+g reduces to 0 and is omitted.

Multiplication — degree increases:

$f(x)g(x) = 30 + 39x + 40x^2 + 17x^3 + 6x^4$ (before reduction)

$f(x)g(x) = 2 + 4x + 5x^2 + 3x^3 + 6x^4$ in $\mathbb{Z}_7[x]$

Multiplying a degree-2 polynomial by a degree-2 polynomial raises the degree to 4.

The Polynomial Ring $\mathbb{Z}_q[x]/(x^n+1)$

To work with polynomials of bounded degree, we perform polynomial arithmetic modulo a fixed polynomial.

Definition (q prime, n a positive integer):

$$R_q = \mathbb{Z}_q[x] / (x^n + 1)$$

Elements of R_q are polynomials of degree less than n, with coefficients in $\mathbb{Z}q$:

$$a_0 + a_1 x + \cdots + a_{n-1} x^{n-1}$$

Two polynomials represent the same element of R_q when their difference is divisible by (x^n+1).

Therefore:

$$x^n \equiv -1 \pmod{x^n + 1}$$

Illustration — $n = 4$: $x^4 \equiv -1$, $x^5 \equiv -x$, $x^6 \equiv -x^2$

Ordinary Polynomial Multiplication

$h(x) = f(x) \cdot g(x)$

(ordinary polynomial product)

Reduction Modulo $(x^n + 1)$

Replace higher powers using $x^n = -1$

(reduces degree, not coefficients)

Result in R_q

Degree < n, coefficients reduced modulo q

(coefficient reduction mod q)

MATHEMATICAL FOUNDATIONS · WORKED EXAMPLE

Polynomial Multiplication and Reduction in Rq

$q = 41, \quad n = 4, \quad R_q = \mathbb{Z}_{41}[x] / (x^4 + 1)$

$f(x) = 32 + 17x + 22x^2, \quad g(x) = 11 + 7x + 19x^2 + x^3, \quad h(x) = f(x) * g(x)$

Step 1 — Multiply in $\mathbb{Z}_{41}[x]$:

$h(x) = 352 + 411x + 969x^2 + 509x^3 + \mathbf{435x^4} + \mathbf{22x^5}$

Step 2 — Reduce modulo $(x^4 + 1)$, using $x^4 = -1$, $x^5 = -x$, $x^6 = -x^2$:

$h(x) = 352 + 411x + 969x^2 + 509x^3 - \mathbf{435} - \mathbf{22x}$

Step 3 — Collect terms and reduce coefficients modulo 41:

$h(x) \equiv -83 + 389x + 969x^2 + 509x^3$

$$= 40 + 20x + 26x^2 + 17x^3$$

$$f(x)g(x) = 40 + 20x + 26x^2 + 27x^3 \quad \text{in } R_q$$

The degree-4, and -5 terms (highlighted) fold back into the constant, x and x^2 coefficients via $x^4 = -1$, $x^5 = -x$.

Representing Polynomials as Vectors

Every element of Rq can be represented by a fixed-length vector of its coefficients.

$f(x) = a_0 + a_1x + \cdots + a_{n-1}x^{n-1}$ corresponds to: $f = (a_0, a_1, \ldots, a_{n-1})$

$f(x) = 23 + 11x^2 + 7x^3$, $g(x) = 40 + 5x + 16x^2$

$f = (23, 0, 11, 7)$ **$g = (40, 5, 16, 0)$**

Vector representation of addition:

$f + g = (63, 5, 27, 7) \bmod 41$

$f + g = (22, 5, 27, 7)$

Coefficient mapping:

Polynomial term	Constant	x	x^2	x^3
f(x)	23	0	11	7
g(x)	40	5	16	0
f(x)+g(x)	**22**	**5**	**27**	**7**